	Attempt	Time	
LATEST	Attempt 1	875 minutes	2 out of 30 *

* Some questions not yet graded

ⓘ Correct answers are hidden.

Score for this quiz: 2 out of 30 *
* Some questions not yet graded
Submitted Nov 8 at 2:38pm
This attempt took 875 minutes.

Question 1
Not yet graded / 2 pts

Question 1: Team & Track Metadata (Fill in the Blank / Short Answer)

Prompt: List the names of all team members. State your project's Analytical Category (Prescriptive, Descriptive, Predictive), Media Type, claimed OSEMN stages, and your fixed Track (ASR or VDB). CIS 731 teams must state their ontology schema version; CIS 531 teams confirm use of the unmodified instructor schema.

Your Answer:

Our team members are Genesis Joseph and Tatenda.

Our project is titled "Analyzing Transaction Data for Fraud Prediction Using Structured Data Analytics."

- **Analytical Category:** Predictive Analytics
- **Media Type:** Structured/Relational Data
- **Taxonomy:** Table C, No. 5
- **OSEMN Stages:** Obtain, Scrub, Explore, Model, and Interpret
- **Fixed Track:** VDB. Our PS4 did not state a track explicitly; we declare VDB because our data is a static, structured corpus, not an audio stream. **[PLEASE CONFIRM DR.HSU]**
- **Ontology Schema Version:** custom ontology `ps6o` v1.0.0, from PS6, with the MP7 extension (`TropeLabel`). That ontology was built for the MP5–MP7 stand-in text corpus, so for the fraud project it will be extended with transaction-level classes (Transaction, Card, Identity, FraudLabel) as version 1.1.0.

Tatenda is responsible for Obtain and Scrub, while Genesis is responsible for Model and Interpret. Our exploratory analysis supports the overall modeling process.

Question 2

Not yet graded / 4 pts

Prompt: Provide your updated problem statement. Have your objectives changed or narrowed since your PS 4 Proposal? List every change with justification, and provide a direct response to every instructor proposal comment.

Your Answer:

Our project's main objective has not changed since PS 4. We are still investigating whether XGBoost can outperform a Logistic Regression baseline in predicting fraudulent transactions using the IEEE-CIS Fraud Detection dataset. We have narrowed our implementation to a reproducible analysis of the original Kaggle files, with a chronological training, validation, and test split.

In response to Dr. Hsu's comments:

1. **Dataset version and provenance:** We obtained the original IEEE-CIS Fraud Detection competition files through the Kaggle API, rather than using a separately transformed Hugging Face dataset. Our original extraction date was October 11, 2026. We recorded file hashes to identify the exact downloaded files. The original source release date still requires confirmation.
2. **Dataset size and schema:** We examined the original transaction and identity tables for both training and testing. The labeled training transaction table contains 590,540 transactions, with approximately 3.5% marked as fraud. The complete table dimensions and hashes are recorded in `results.json`.
3. **Taxonomy:** We confirmed that the project is Predictive Analytics using Structured/Relational Data, corresponding to Table C, No. 5.
4. **OSEMN plan:** Obtain covers Kaggle API ingestion and source documentation. Scrub covers type validation, missing-value handling, transaction ID checks, and duplicate detection. Explore covers missingness, fraud-rate drift, transaction amounts, and UMAP projections. Model compares Logistic Regression against XGBoost. Interpret evaluates performance, errors, and feature behavior.
5. **Data validity and cleaning:** We preserve missing information instead of automatically removing rows. We check transaction IDs, target labels, and numeric data types. Invalid records are flagged for investigation to avoid unnecessarily removing fraudulent transactions.
6. **Modeling platform and ground truth:** We use Python and scikit-learn for our preliminary Logistic Regression baseline and plan to use XGBoost for comparison. The ground-truth fraud labels come from the `isFraud` column in the original Kaggle training data.
7. **Exploration and model selection:** Our figures show substantial missingness, changing fraud rates over time, and overlapping transaction amounts between fraud and non-fraud cases. These observations support our missing-value policy, chronological validation strategy, and use of multiple features for classification.

8. **Notebook and evaluation:** We created a reproducible analysis notebook and established a preliminary Logistic Regression baseline. We reserved the newest 20% of the labeled data for final evaluation and will select models using the validation period rather than the held-out test period.

The main changes since the proposal are more detailed provenance records, implemented exploratory analysis, an initial measured baseline, and a documented evaluation strategy. The research objective and proposed model comparison remain unchanged.

Question 3
Not yet graded / 8 pts

Prompt: Provide your data provenance table (corpus/stream source, whitelist ID, size, ingest method; confirming compliance with restricted topics and terms of use license). Include descriptions and findings for at least three exploratory figures, including a UMAP sensitivity sweep over `n_neighbors` and `min_dist` (with fixed `random_state` following Wilke Parts I–II) and a cohort/drift plot with uncertainty. Report your initial advisory performance metric on your naive baseline.

Your Answer:

We used the original IEEE-CIS Fraud Detection dataset from Kaggle, downloaded through the Kaggle API on October 11, 2026. We recorded file sizes and SHA-256 hashes so we can identify the exact files used in our analysis.

Data provenance and table sizes

File	Rows	Columns	Size (MB)
train_transaction.csv	590,540	394	683.35
train_identity.csv	144,233	41	26.53
test_transaction.csv	506,691	393	613.19
test_identity.csv	141,907	41	25.80
Total	**1,383,371**	—	**1,348.87**

The labeled training transaction table contains 20,663 fraudulent transactions, or 3.499%. Identity records cover approximately 24.42% of training transactions. The transaction timestamps span approximately 182 days.

Our data-validity checks passed: transaction IDs were unique and non-null, fraud labels were limited to 0 and 1, transaction amounts were positive, and identity records followed a one-to-one relationship with transaction IDs.

Exploratory figures and findings

1. **Missingness:** Several feature groups have substantial missing values. The highest missingness rates include R_email (76.75%), dist (76.64%), D (58.15%), M (49.92%), and V (43.04%). We plan to

preserve missingness information rather than automatically delete transa

2. **Fraud-rate drift:** We examined 27 weeks of transactions. Weekly fraud rates ranged from 1.85% to 5.06%, increasing from 2.81% in the first week to 4.32% in the last week. Our weekly drift plot includes 95% confidence intervals. This supports using chronological splits to evaluate performance on newer transactions.
3. **Transaction amounts:** The median amount was $68.50 for non-fraudulent transactions and $75.00 for fraudulent transactions. The amounts overlap, suggesting that transaction amount alone is not enough to identify fraud reliably.
4. **UMAP sensitivity:** We created nine UMAP projections to examine how the visual structure changes with different `n_neighbors` and `min_dist` settings. These figures are exploratory and are not used as proof of classification performance.

Preliminary baseline and validation

We split the 590,540 labeled transactions chronologically into 413,378 training rows (70%), 59,053 validation rows (10%), and 118,109 held-out test rows (20%).

The validation fraud rate was 3.4901%. The majority-class baseline had a PR-AUC of 0.034901 and ROC-AUC of 0.5000.

Our Logistic Regression model used 211 features and balanced class weights. It achieved:

- **PR-AUC:** 0.379453
- **ROC-AUC:** 0.844564

This gives us a preliminary baseline for our planned XGBoost comparison. All reported model scores are advisory validation results, not final test results. The held-out test set has not been used for final evaluation.

Source and compliance

We used the original Kaggle competition files rather than an externally transformed Hugging Face dataset. Our `results.json` contains the exact SHA-256 hashes for all four files. We still need to document the source whitelist ID, applicable data-use terms, and confirmation of course compliance.

For our UMAP sensitivity analysis, we used 6,000 sampled transactions and 208 features. We tested `n_neighbors` values of 5, 15, and 50 with `min_dist` values of 0.0, 0.1, and 0.5, producing nine configurations. We fixed `random_state=42` for reproducibility.

Our weekly fraud-rate analysis covered 27 weeks and included 95% Wilson confidence intervals to show uncertainty in the estimated fraud rates.

Question 4

Not yet graded / 5 pts

Prompt: Detail your pipeline architecture (ingest, embed/transcribe, index, retrieve/analyze, evaluate) and explain why you chose this approach in alignment with your OSEMN stages. Provide your component-version table that you will include in your final project deliverable as a **fixed project manifest (https://k-state.instructure.com/courses/201513/pages/term-project-deliverable-how-to)** and cite your data source and overall objective to allow enforced requirements to be checked.

Your Answer:

Our PS 8 analysis uses Python to examine the original IEEE-CIS Fraud Detection dataset from Kaggle. The code loads the training transaction and identity CSV files, checks the data, creates exploratory figures, prepares features, and evaluates a Logistic Regression baseline.

Pipeline architecture

1. **Data loading:** The `load_train()` function reads `train_transaction.csv` and `train_identity.csv` using pandas. It uses `float32` for numerical transaction features to reduce memory usage. The `data_card()` function records row counts, column counts, file sizes, and SHA-256 hashes for all four main CSV files.
2. **Data validation:** The code checks whether transaction IDs are unique and non-null, fraud labels are 0 or 1, transaction amounts are positive, and identity transaction IDs are unique. All five recorded checks passed. The analysis does not automatically delete rows because of missing values.
3. **Exploratory analysis:** The code generates four figures: missingness by feature group, weekly fraud rates with 95% Wilson confidence intervals, transaction amounts by fraud class, and a UMAP sensitivity sweep. The UMAP analysis uses 6,000 sampled transactions, 208 features, nine parameter combinations, and `random_state=42`.
4. **Feature preparation:** The `numeric_features()` function selects numerical features with no more than 50% missing values. The baseline pipeline replaces missing numerical values with the median and standardizes them. For `ProductCD`, `card4`, and `card6`, it replaces missing values with the category "missing" and applies one-hot encoding.
5. **Modeling and evaluation:** The code divides the labeled transactions chronologically into 413,378 training rows, 59,053 validation rows, and 118,109 held-out test rows. Logistic Regression uses balanced class weights and 211 selected input features. It achieved a validation PR-AUC of 0.379453 and ROC-AUC of 0.844564. The majority-class baseline achieved a PR-AUC of 0.034901 and ROC-AUC of 0.500000.

Component-version manifest

Component	Recorded version

Python	3.12.1
Operating system	macOS 26.6.2, ARM64
pandas	3.1.1
NumPy	2.5.3
scikit-learn	1.9.1
umap-learn	0.5.12
Matplotlib	3.11.2
Kaggle API	2.2.5
XGBoost	Not installed
Random state	42

Our source is the original IEEE-CIS Fraud Detection Kaggle competition dataset. The objective recorded in our project proposal is to predict fraudulent transactions and compare XGBoost with a Logistic Regression baseline.

The current code implements the exploratory analysis and Logistic Regression baseline. It saves the figures in `interim/figures/` and the recorded results in `interim/results.json`. All reported model metrics are from the validation period, and `grading_eligible` is set to `false`.

Question 5
Not yet graded / 5 pts

Prompt: Provide at least 5 quantified risks (likelihood, impact, trigger, and mitigation)—for example, running out of API credits or exceeding your local compute limits, with a CPU-baseline fallback plan ready. Outline your work breakdown and your specific data annotation plan. This must include your 50–200 labeled example target, how you are sourcing pre-annotated data or using automated analytical preprocessing to scale and pre-label unlabeled data into a silver standard, and your **Inter-Annotator Agreement (IAA)** protocol (such as multi-rater consistency metrics).

Finally, explain your validation strategy, including why keeping a strict **fractional cap on your validation set** (such as capping validation splits under 1% when working with large scaled datasets) acts as a vital safety criterion to prevent data leakage, overfitting, and evaluation noise.

> **Authoritative References & Tutorials:**
>
> - **Jurafsky & Martin, *Speech and Language Processing (3rd ed. draft)* — Text Classification, Silver Standards, and Inter-Annotator Agreement (https://web.stanford.edu/~jurafsky/slp3/)**
> - **James, Witten, Hastie, Tibshirani, & Taylor (2023), *An Introduction to Statistical Learning (ISLP)* — Chapter 5 (Validation Strategy, Data Splits, and Resampling Methods) (https://www.statlearning.com/)**

Your Answer:

1. Five quantified project risks

Your dataset contains 590,540 labeled training transactions, 20,663 fraud cases (3.50%), and 27 weeks of data. Your Logistic Regression baseline achieved a validation PR-AUC of 0.379453.

The risk likelihoods and impact ratings below are suggested assessments, not measured probabilities.

Risk	Likelihood / Impact (proposed)	Trigger to monitor	Mitigation
Missing data	High / High	A feature group exceeds 50% missingness	Keep raw data, document missingness, use imputation for the baseline, and compare missingness-aware alternatives
Temporal drift	High / High	Weekly fraud rate moves by more than 2 percentage points from the training-period rate	Compare performance across time periods and retain chronological splits
Class imbalance	High / High	Validation PR-AUC falls below 0.20	Use class weighting, monitor precision and recall, and compare against the 0.034901 prevalence baseline
Memory limitations	Medium / Medium	A process runs out of memory or UMAP exceeds 30 minutes	Continue using `float32`, sampled UMAP, and a CPU-compatible smaller-feature baseline
Data quality or leakage	Medium / High	Any invalid ID, invalid target, duplicate join, or held-out test access during tuning	Stop the affected pipeline, inspect the records, correct the processing logic, and repeat validation

The 50% missingness threshold is already reflected in your numerical feature selection. The 2-percentage-point drift threshold, 0.20 PR-AUC threshold, and 30-minute runtime limit are proposals, not values established by your code.

I would also include a sixth risk if space permits: dataset licensing or course approval, triggered if the Kaggle terms or course whitelist cannot be confirmed. This was an explicit concern in Dr. Hsu's feedback.

2. Work breakdown and deadlines

Your final deadline is December 14, 2026. I recommend finishing the main work by December 7, leaving seven days for corrections.

Nov 9–15

Tatenda

Finalize data provenance, cleaning rules, validity checks, and course licensing evidence.

Nov 9–15

Genesis

Finalize baseline documentation, feature processing, and evaluation plan.

Nov 16–22

Both

Select and independently review 120 transactions; compare annotations.

Nov 23–29

Genesis

Implement and evaluate XGBoost against Logistic Regression using validation data.

Nov 23–29

Tatenda

Document cleaned-data schema, missingness decisions, and data-quality findings.

Nov 30–Dec 6

Both

Complete error analysis, final evaluation when the models are frozen, report, and presentation.

Dec 7–13

Both

Buffer for corrections, reproducibility checks, instructor feedback, and submission preparation.

All dates are my proposed schedule, not previously agreed team commitments. This plan assumes both of you can work during the remaining weeks.

3. How many transactions should you check?

Recommended annotation sample

120 transactions

Both teammates independently review the same 120 records.

I suggest selecting the sample from your training-period data only, so the annotation exercise does not influence the held-out test set.

Sample group	Number	Selection method
Known fraud cases	40	Random sample from training records with `isFraud=1`
Known non-fraud cases	40	Random sample from training records with `isFraud=0`
Difficult examples	40	Select using model scores: 20 near the proposed decision threshold, 10 high-score and 10 low-score cases
Total	120	

This is a proposed stratified review sample, not a representative estimate of the overall 3.5% fraud prevalence.

A major limitation: transaction features alone generally cannot establish whether fraud actually occurred. You should use the existing Kaggle `isFraud` labels as the ground truth and have reviewers annotate what they can genuinely assess, such as record validity, suspicious patterns, or whether the available evidence is insufficient. Don't claim the two reviewers independently verified actual fraud without external investigation evidence.

4. Using model scores to select examples

Your Logistic Regression produces scores that can help identify records worth reviewing.

For example, using a proposed 0.50 threshold:

Illustrative model-score groups

0.05

Low score

0.49

Near threshold

0.95

High score

These scores are examples, not actual transactions from your dataset. The 0.50 threshold is illustrative and has not been selected or validated.

You could select uncertain examples close to the threshold and high-confidence examples at either extreme.

However, a model score is not a label. A score of 0.95 does not prove fraud, and your balanced Logistic Regression scores should not automatically be interpreted as calibrated fraud probabilities.

For your silver-standard plan, I recommend keeping model-generated suggestions in a separate field such as `model_suggestion`, with `source="model"` and `verified=false`. Only after a defined review process should a suggested annotation be accepted. The original `isFraud` label remains separate and unchanged.

5. How to calculate agreement between you and Tatenda

I recommend using Cohen's kappa for a clearly defined binary review question, such as whether a transaction passes your documented data-quality review.

Both reviewers should examine the same 120 transactions independently, without seeing each other's answers or model scores.

For illustration, suppose your results were:

	Tatenda: Flag	**Tatenda: No flag**
Genesis: Flag	40	10
Genesis: No flag	5	65

In this hypothetical example, you agree on 105 of 120 records (87.5%), and Cohen's kappa is approximately 0.74 after accounting for agreement expected by chance.

Proposed agreement targets

κ ≥ 0.60 — more permissive

κ ≥ 0.70 — suggested target
κ ≥ 0.80 — stricter target

These are proposed acceptance criteria, not course-prescribed thresholds.

My suggestion is κ ≥ 0.70 as an initial team target. If agreement is lower, review disagreements, improve the annotation instructions, and conduct another independent review. Do not silently change the original labels just to increase agreement.

If your labels are all identical, kappa can be undefined, so also report the label distribution and raw agreement.

6. Why keep 10% for validation?

Your actual split is:

Recorded chronological split

Training · 70%

413,378

Validation · 10%

59,053

Held-out · 20%

118,109

Your validation fraud rate is 3.4901%. That means approximately 2,061 fraud cases in the 10% validation period, compared with approximately 206 fraud cases in a 1% sample at the same prevalence.

The larger validation period gives you more positive cases to estimate precision-recall performance, especially when fraud is rare. It also preserves a meaningful consecutive time period for examining changing fraud patterns.

However, a 10% validation set does not automatically prevent overfitting, and a 1% set does not automatically cause it. Overfitting depends on how often and how extensively you use validation feedback to make decisions.

I suggest these safeguards:

- Use only the training period to fit preprocessing steps and models.
- Limit and document hyperparameter experiments.
- Select the final model and threshold using validation data only.
- Avoid changing the model repeatedly after reviewing validation performance.

- Keep the newest 20% untouched until the final model is frozen.

Ready-to-use Question 5 draft

This draft separates completed analysis from the plan you still need to approve.

Question 5 — Risk Register and Revised Plan

Our analysis identified five main risks: missing data, temporal drift, class imbalance, memory limitations, and data-quality errors.

Missing data is a high-likelihood, high-impact risk. Some feature groups exceed 50% missingness, including R_email (76.75%) and dist (76.64%). We preserve the original records and handle missing values during feature preparation.

Temporal drift is also a high-likelihood, high-impact risk. Weekly fraud rates ranged from 1.85% to 5.06% across 27 weeks. We use chronological data splits and propose monitoring changes greater than two percentage points.

Class imbalance is a high-likelihood, high-impact risk because only 20,663 of 590,540 training transactions are fraudulent (3.50%). Our Logistic Regression baseline uses balanced class weights. We propose investigating a validation PR-AUC below 0.20, compared with our current baseline of 0.379453.

Memory limitations are rated medium likelihood and medium impact. The four CSV files total approximately 1.35 GB. Our code uses float32 for numerical features and samples 6,000 transactions for UMAP. We propose switching to smaller CPU-based experiments if memory or runtime limits are reached.

Data-quality errors are rated medium likelihood and high impact. Our code checks transaction IDs, fraud labels, positive amounts, and identity relationships. All five recorded checks passed. Any failed check in future processing would trigger investigation before modeling.

For our revised plan, we propose completing data documentation, model development, and annotation review during November, finalizing evaluation and reporting by December 6, and reserving December 7–13 as a buffer before the December 14 deadline.

We propose reviewing 120 training-period transactions, including 40 known fraud cases, 40 known non-fraud cases, and 40 examples selected using model scores. Both team members would independently review the same records using a written annotation guide. Model scores would only help select examples and would not replace the original Kaggle fraud labels.

We propose using Cohen's kappa to measure agreement on a clearly defined review label, with an initial target of at least 0.70. Disagreements would be reviewed and documented. This annotation and agreement process is planned and has not yet been performed.

Our existing validation split uses 59,053 transactions (10%) and leaves 118, sactions (20%) for held-out testing. We retain the larger validation period because fraud is rare, and a 1% sample would contain far fewer positive examples. To reduce overfitting, we will limit repeated model tuning, fit preprocessing only on training data, and avoid using the held-out test set for model selection.

The risk ratings, proposed thresholds, annotation sample, agreement target, and remaining deadlines are team planning decisions rather than completed experimental results.

Question 6
2 / 2 pts

Prompt: I attest that the written responses provided in this survey originate from my own thoughts, code, and bullet points. Where Generative AI was used, it was restricted to the role of an editor/synthesizer in accordance with the course GenAI Kit (v1-3). I did not use prompts that delegated the entire cognitive load to an AI system.

- (•) True
- ( ) False

Question 7
Not yet graded / 2 pts

Prompt: Upload your `GenAI.pdf` audit log here. This document must contain your exact prompts used during this interim phase, your original draft text/bullet points that you fed into the AI, and the AI's raw output.

(If you did not use GenAI for any part of PS 8, upload a PDF explicitly stating "No Generative AI was used for PS 8").

GenAi8.pdf (https://k-state.instructure.com/files/45696387/download)

Question 8
Not yet graded / 2 pts

Prompt: Provide the URL to your team's active GitHub repository. The repository must show a commit history demonstrating your working code skeleton and baseline implementation under the `ps8-submission` tag (with figure sources in `interim/`).

Your Answer:

https://github.com/genesisJoseph/cis531-731-2026_fall/tree/ps8-submission

Score: 2 out of 30

* Some questions not yet graded